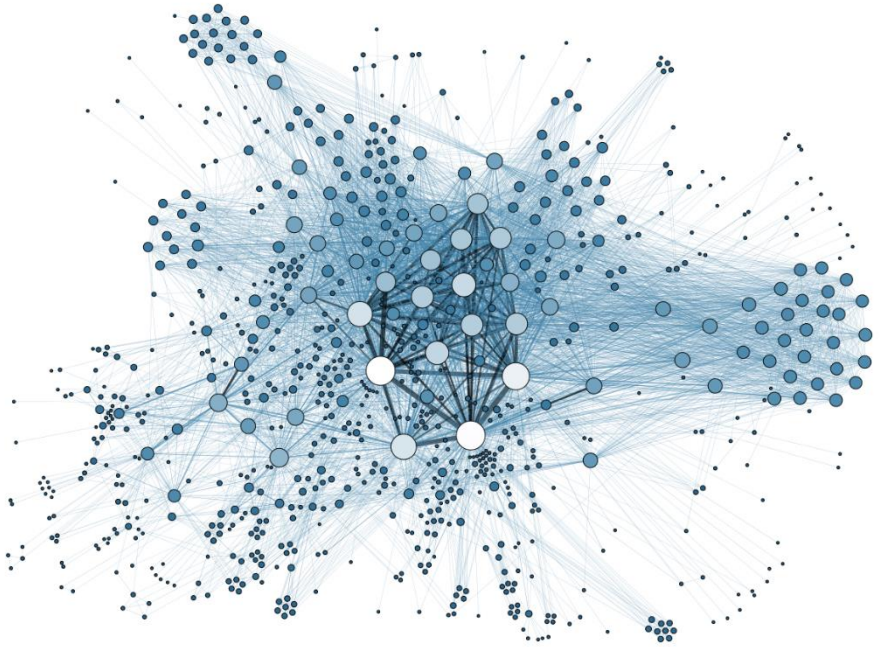


Information Diffusion



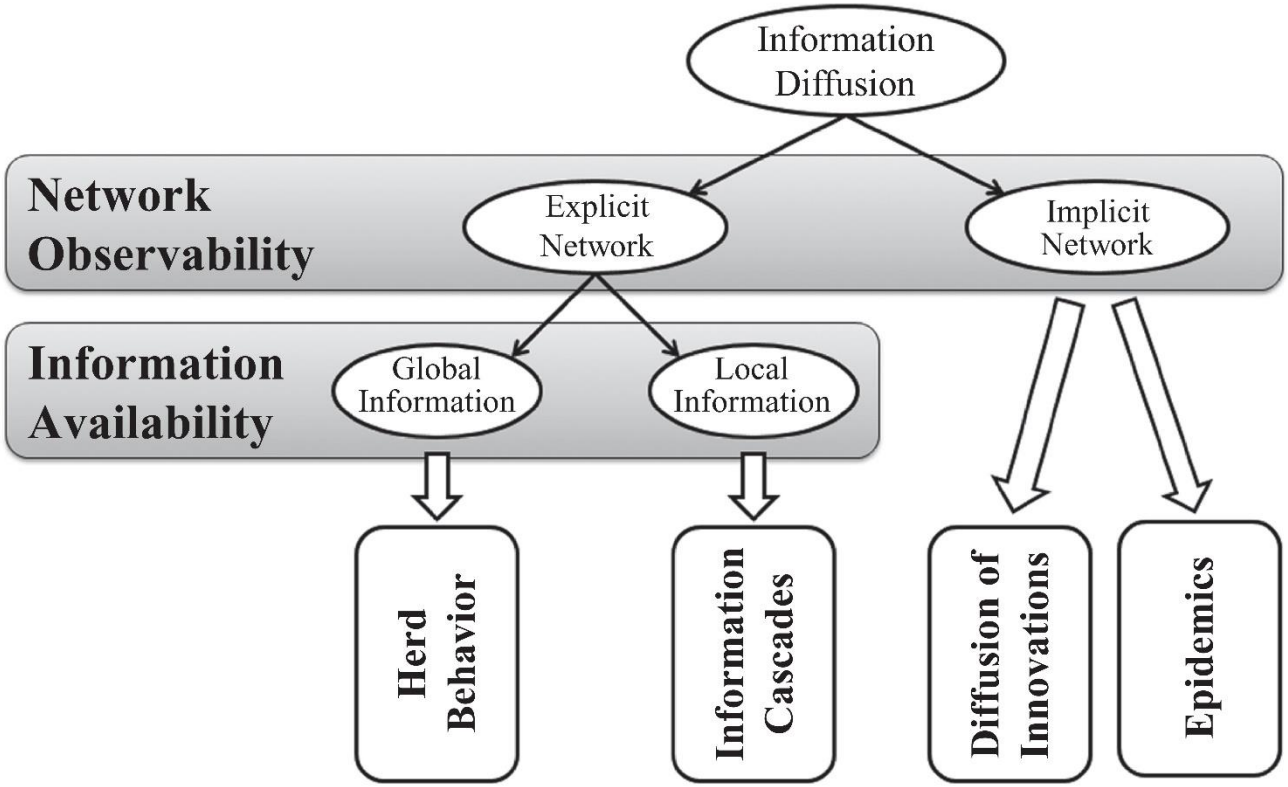
Information Diffusion

- Information diffusion: process by which a piece of information (knowledge) is spread and reaches individuals through interactions.

Information Diffusion

- **Sender(s).** A sender or a small set of senders that initiate the information diffusion process;
- **Receiver(s).** A receiver or a set of receivers that receive diffused information. Commonly, the set of receivers is much larger than the set of senders and can overlap with the set of senders; and
- **Medium.** This is the medium through which the diffusion takes place. For example, when a rumor is spreading, the medium can be the personal communication between individuals

Information Diffusion Types



We define the process of interfering with information diffusion by expediting, delaying, or even stopping diffusion as Intervention

Herd Behavior

- Network is observable
- Only public information is available

Herd Behavior: Popular Restaurant Example

- Assume you are on a trip in a metropolitan area that you are less familiar with.
- Planning for dinner, you find restaurant **A** with excellent reviews online and decide to go there.
- When arriving at **A**, you see **A** is almost empty and restaurant **B**, which is next door and serves the same cuisine, almost full.
- Choosing Restaurant **B** over Restaurant **A** because it is crowded—despite A having better online reviews—is an example of herd behaviour.

Herd Behavior: Milgram's Experiment

Stanley Milgram asked one person to stand still on a busy street corner in New York City and stare straight up at the sky

- About 4% of all passersby stopped to look up.



When 5 people stand on the sidewalk and look straight up at the sky, 20% of all passersby stopped to look up.

Finally, when a group of 18 people look up simultaneously, almost 50% of all passersby stopped to look up.

Herd Behavior

Herd behavior describes when a group of individuals performs actions that are highly correlated without any plans

Main Components of Herd Behavior

- Connections between individuals
- A method to transfer behavior among individuals or to observe their behavior

Examples of Herd Behavior

- Flocks, herds of animals, and humans during sporting events, demonstrations, and religious gatherings

Network Observability in Herb Behavior

In herd behavior, individuals make decisions by **observing all other individuals' decisions**

- In general, herd behavior's network is close to a **complete graph** where nodes can observe at least most other nodes and they can observe **public information**
  - For example, they can see the crowd

Designing a Herd Behavior Experiment

1. There needs to be a decision made.
   - In our example, it is going to a restaurant
2. Decisions need to be in sequential order
3. Decisions are not mindless and people have private information that helps them decide
4. No message passing is possible. Individuals don't know the private information of others, but can infer what others know from what is observed from their behavior.

Herding Intervention

Herding: we only have access to public information

- Herding may be intervened by releasing private information which was not accessible before

Herding Intervention

To intervene the herding effect, we need one person to tell the herd that there is nothing in the sky and the first person doing this to stop his nose bleeding



How Does Intervention Work?

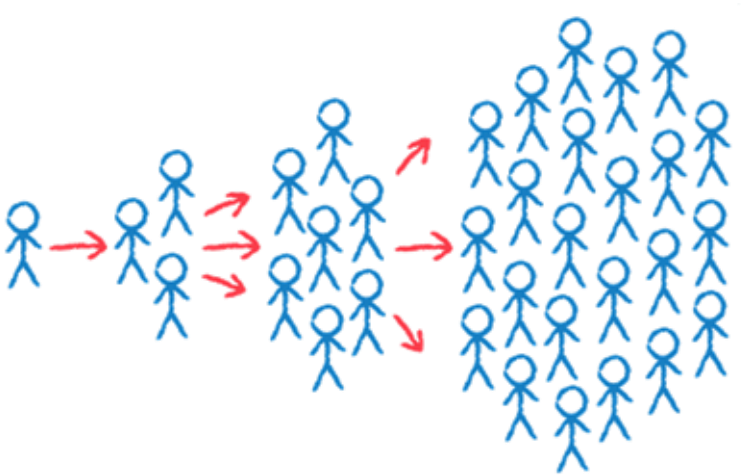
- When a new piece of private information releases,
  - The herd reevaluate their guesses and this may create completely new results

Information Cascade

- In the presence of a network
- Only local information is available

Information Cascade

- Users often repost content posted by others in the network.
  - Content is often received via immediate neighbors (friends).





Information propagates through friends

An information cascade: a piece of information/decision cascaded among some users, where

individuals are connected by a network and

individuals are only observing decisions of their immediate neighbors (friends).

Cascade users have less information available

- Herding users have almost all information about decisions

Underlying Assumptions for Cascade Models

- The network is a **directed graph**.
  - Nodes are actors
  - Edges depict the communication channels between them.
- A node can only influence nodes that it is connected to
- Decisions are binary. nodes can
  - **Active**: the node has adopted the behavior/innovation/decision;
  - **Inactive**
- A activated node can activate its neighboring nodes; and
- Activation is a progressive process, where nodes change from inactive to active, but not vice versa

Independent Cascade Model (ICM)

- **Independent Cascade Model (ICM)**
  - Sender-centric model of cascade
  - Each node has **one chance** to activate its neighbors

- In ICM, nodes that are active are senders and nodes that are being activated as receivers

ICM Algorithm

- Node activated at time t, has one chance, at time step $t + 1$, to activate its neighbors
- Assume v is activated at time t
  - For any neighbor w of v, there's a probability p_{vw} that node w gets activated at time $t + 1$.
- Node v activated at time t has a single chance of activating its neighbors
  - The activation can only happen at $t + 1$

ICM Algorithm

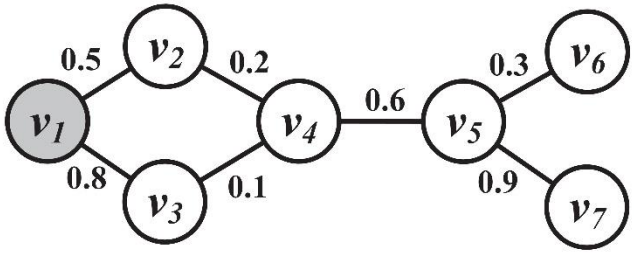
Algorithm 1 Independent Cascade Model (ICM)

Require: Diffusion graph $G(V, E)$, set of initial activated nodes A_0, activation probabilities $p_{v,w}$

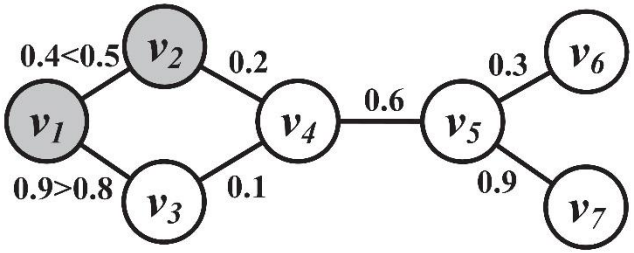
return Final set of activated nodes A_∞
i = 0;
while $A_i \neq \{\}$ **do**

  $i = i + 1$;
  $A_i = \{\}$;
  for all $v \in A_{i-1}$ **do**
    for all w neighbor of $v, w \notin \cup_{j=0}^{i} A_j$ **do**
      rand = generate a random number in [0,1];
      if rand $< p_{v,w}$ **then**
        activate w;
        $A_i = A_i \cup \{w\}$;
      end if
    end for
  end for
end while
$A_\infty = \cup_{j=0}^{i} A_j$;
Return A_∞;

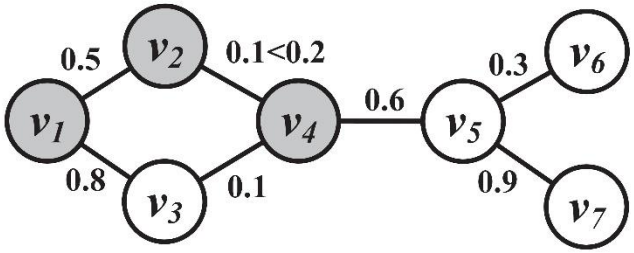
Independent Cascade Model: An Example



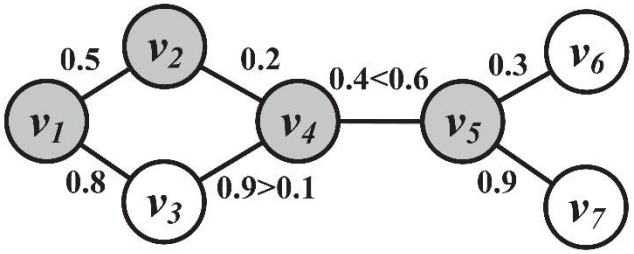
Step 1



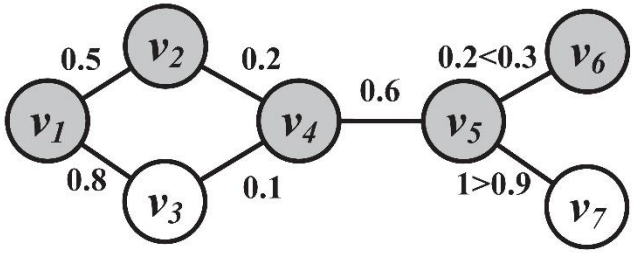
Step 2



Step 3



Step 4



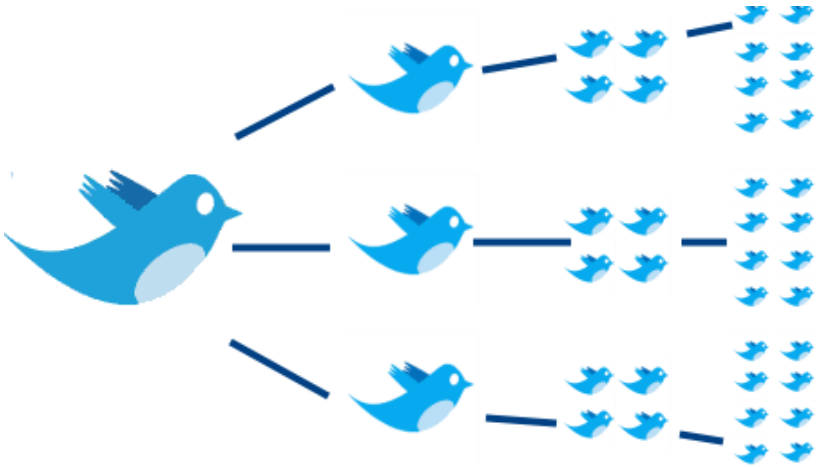
Step 5

Maximizing the Spread of Cascades

Maximizing the spread of cascades

- **Maximizing the Spread of Cascades** is the problem of finding a small set of nodes in a social network such that their aggregated spread in the network is maximized

- Applications
  - Product marketing
  - Influence

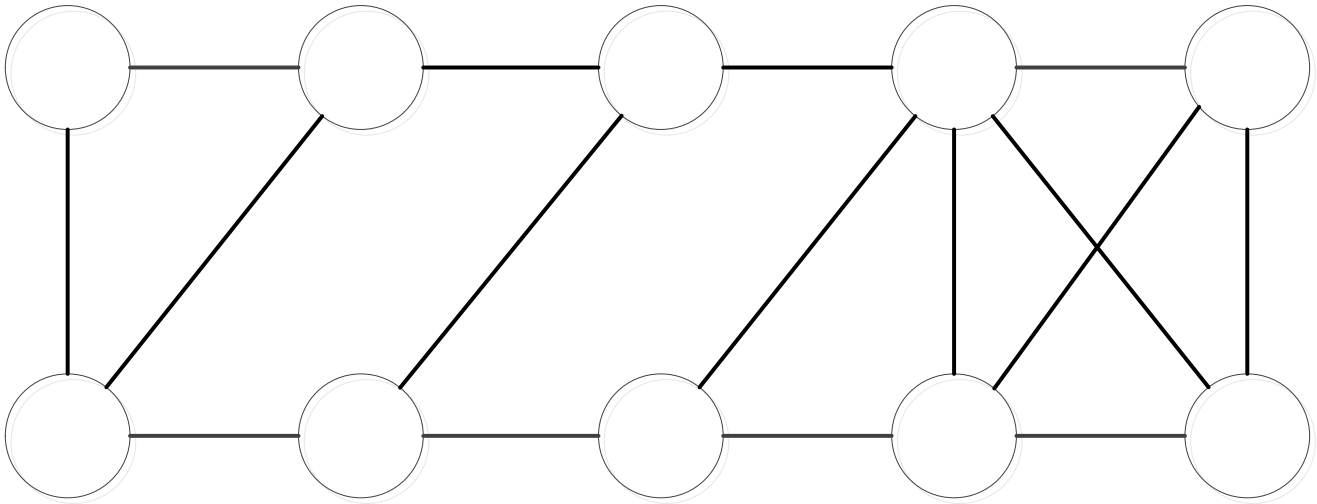


Problem Setting

- **Given**
  - A limited budget **B** for initial advertising
    - Example: give away free samples of product
  - Estimating spread between individuals
- **Goal**
  - To trigger a large spread
    - i.e., further adoptions of a product
- **Question**
  - Which set of individuals should be targeted at the very beginning?

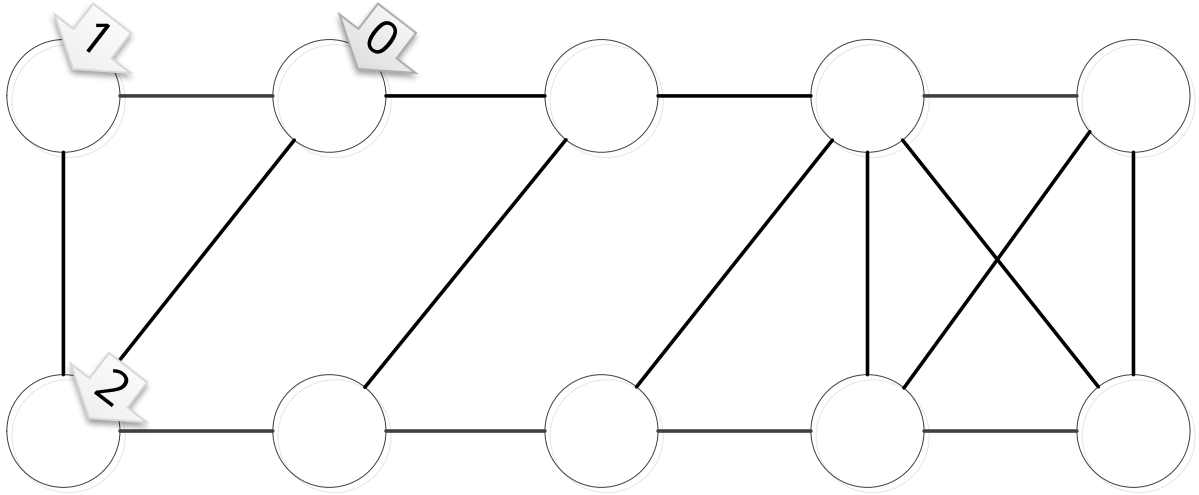
Maximizing the Spread of Cascade: Example

- We need to pick k nodes such that maximum number of nodes are activated

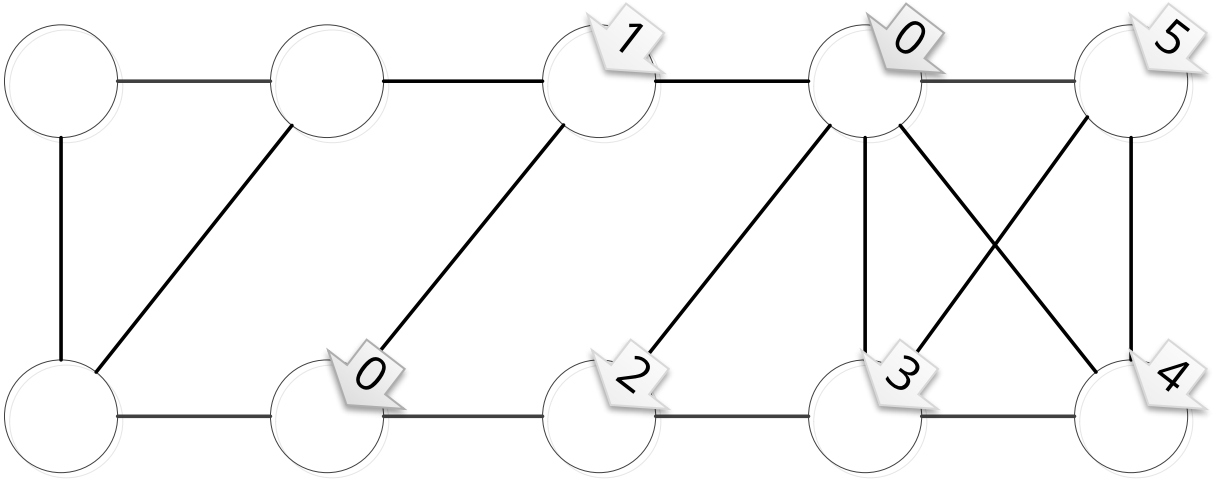


Maximizing the Spread of Cascade

Select one seed



Select two seeds



Problem Statement

- Spread of node set S: $f(S)$
  - An expected number of activated nodes at the end of the cascade, if set S is the initial active set

- Problem:
  - Given a parameter k (budget), find a k-node set S to maximize $f(S)$
  - A constrained optimization problem with $f(S)$ as the objective function

f(S): Properties

1. **Non-negative** (obviously)

2. **Monotone**

$$f(S+v) \geq f(S)$$

3. **Submodular**
   - Let N be a finite set
   - A set function is submodular if and only if

$$f: 2^N \text{ a } \Re$$

$$\forall S \subset T \subset N, \forall v \in N \setminus T,$$

$$f(S+v) - f(S) \geq f(T+v) - f(T)$$

Some Facts Regarding this Problem

- **Bad News**
  - Consider a non-negative, monotone, submodular function f
  - Finding a k-element set S for which $f(S)$ is maximized is **NP-hard**
    - It is NP-hard to determine the optimum initial set for cascade maximization
- **Good News**
  - We can use a greedy algorithm
    - Start with an empty set S
    - For k iterations:

      Add node v to S that maximizes $f(S \cup \{v\}) - f(S)$.

  - How good (or bad) it is? (Kempe et al., before that Nemhauser et al.)
    - **Theorem**: the greedy algorithm provides a $(1 - 1/e)$ approximation.
    - The resulting set S activates **at least** $(1 - 1/e) \approx 63\%$ of the number of nodes that any size-k set S could activate.

Greedy Algorithm

Algorithm 1 Maximizing the spread of cascades – Greedy algorithm

Require: Diffusion graph $G(V, E)$, budget k

1: **return** Seed set S (set of initially activated nodes)
2: $i = 0$;
3: $S = \{\}$;
4: **while** $i \neq k$ **do**
5: $v = \arg\max_{v \in V \setminus S} f(S \cup \{v\})$;
`or equivalently` $\arg\max_{v \in V \setminus S} f(S \cup \{v\}) - f(S)$
6: $S = S \cup \{v\}$;
7: $i = i + 1$;
8: **end while**
9: Return S;

Consider a network $G(V, E)$ with 5 nodes:

$$V = \{A, B, C, D, E\}$$

Suppose the reach/cascade function $f(S)$ represents the total number of unique nodes reached (including the seed nodes themselves) under a simple deterministic reachability model:

- **Node** A is connected to $\{B, C\}$. Reaching A activates $\{A, B, C\}$.
- **Node** B is connected to $\{C\}$. Reaching B activates $\{B, C\}$.
- **Node** C is connected to no outgoing nodes. Reaching C activates $\{C\}$.
- **Node** D is connected to $\{E\}$. Reaching D activates $\{D, E\}$.
- **Node** E is connected to no outgoing nodes. Reaching E activates $\{E\}$.

2. Target Budget

Set the budget to $k = 2$.

Execution Steps
Initialization (Lines 2–3)
•Set iteration counter i=0
•Initialize seed set S={}
•Initial spread function value f(∅)=0

Iteration 1 (i=0, target k=2)
Evaluate the marginal gain $f(S \cup \{v\}) - f(S)$ for every node $v \in V \setminus S = \{A,B,C,D,E\}$:

- For $v = A$: $S \cup \{A\} = \{A\} \rightarrow \text{reaches } \{A, B, C\} \implies f(\{A\}) = 3$

  Marginal gain: $f(\{A\}) - f(\emptyset) = 3 - 0 = 3$
- For $v = B$: $S \cup \{B\} = \{B\} \rightarrow \text{reaches } \{B, C\} \implies f(\{B\}) = 2$

  Marginal gain: $f(\{B\}) - f(\emptyset) = 2 - 0 = 2$
- For $v = C$: $S \cup \{C\} = \{C\} \rightarrow \text{reaches } \{C\} \implies f(\{C\}) = 1$

  Marginal gain: $f(\{C\}) - f(\emptyset) = 1 - 0 = 1$
- For $v = D$: $S \cup \{D\} = \{D\} \rightarrow \text{reaches } \{D, E\} \implies f(\{D\}) = 2$

  Marginal gain: $f(\{D\}) - f(\emptyset) = 2 - 0 = 2$
- For $v = E$: $S \cup \{E\} = \{E\} \rightarrow \text{reaches } \{E\} \implies f(\{E\}) = 1$

  Marginal gain: $f(\{E\}) - f(\emptyset) = 1 - 0 = 1$

- **Selection (Lines 5–7):**
- $v = \text{argmax}_{v \in V \setminus S}(f(S \cup \{v\}) - f(S)) = A$ (highest marginal gain of 3)
- Update seed set: $S = \{\} \cup \{A\} = \{A\}$
- Update counter: $i = 0 + 1 = 1$

Iteration 2 (i=1, target k=2)

- Current seed set S={A}, with total reach f({A})=3 ({A,B,C}).
- Evaluate remaining candidate nodes v∈V\S={B,C,D,E}:

- For $v = B$: $S \cup \{B\} = \{A, B\} \rightarrow$ reaches $\{A, B, C\} \implies f(\{A, B\}) = 3$

  Marginal gain: $f(\{A, B\}) - f(\{A\}) = 3 - 3 = 0$
- For $v = C$: $S \cup \{C\} = \{A, C\} \rightarrow$ reaches $\{A, B, C\} \implies f(\{A, C\}) = 3$

  Marginal gain: $f(\{A, C\}) - f(\{A\}) = 3 - 3 = 0$
- For $v = D$: $S \cup \{D\} = \{A, D\} \rightarrow$ reaches $\{A, B, C, D, E\} \implies f(\{A, D\}) = 5$

  Marginal gain: $f(\{A, D\}) - f(\{A\}) = 5 - 3 = 2$
- For $v = E$: $S \cup \{E\} = \{A, E\} \rightarrow$ reaches $\{A, B, C, E\} \implies f(\{A, E\}) = 4$

  Marginal gain: $f(\{A, E\}) - f(\{A\}) = 4 - 3 = 1$

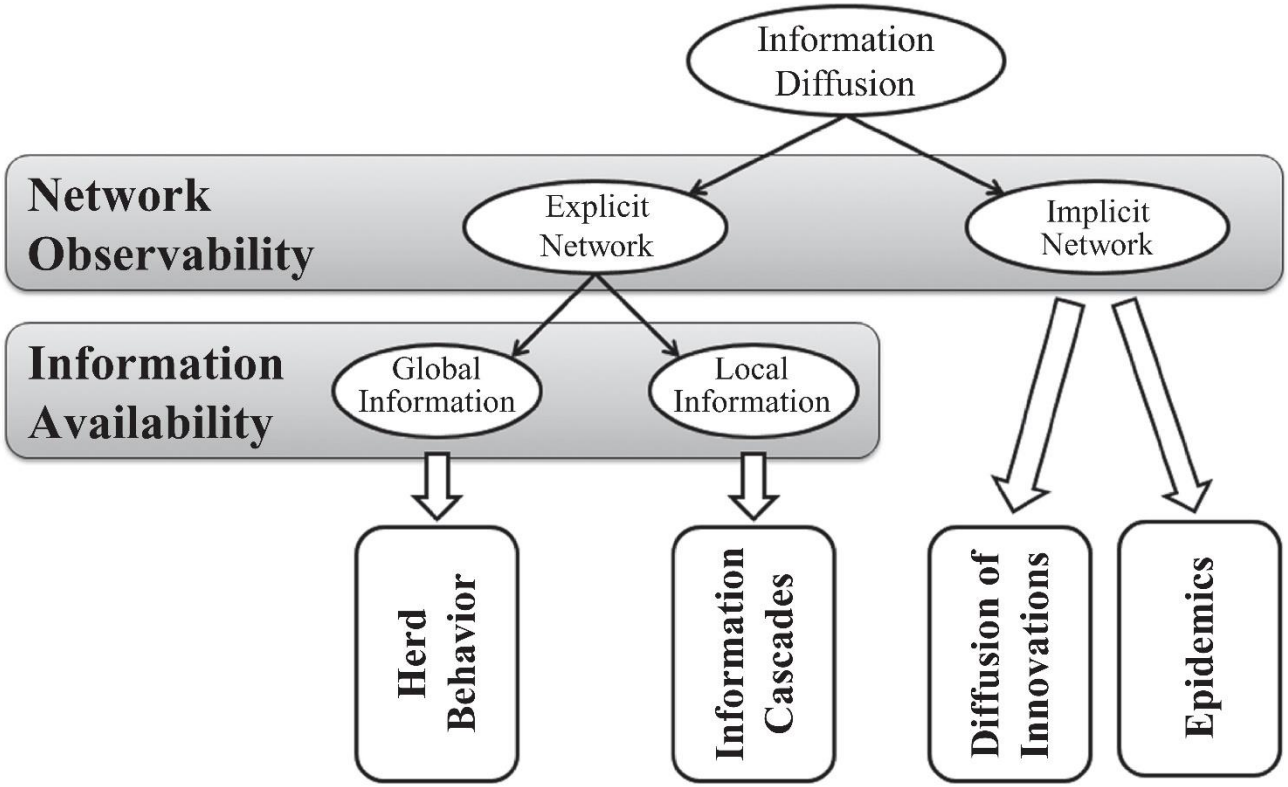
Selection (Lines 5–7):
•v=argmaxv∈V\S(f(S∪{v})−f(S))=D (highest marginal gain of 2)
•Update seed set: S={A}∪{D}={A,D}
•Update counter: i=1+1=2

Return Seed Set: S={A,D}
Total Cascade Spread Achieved: f({A,D})=5 (all nodes reached).

Intervention

- By limiting the number of out-links
  - Disconnected nodes don't get to activate anyone
- By limiting the number of in-links
  - Reducing the chance of getting activated by others
- By decreasing the activation probability p_{vw}
  - Reducing the chance of activating others.

Information Diffusion Types



We define the process of interfering with information diffusion by expediting, delaying, or even stopping diffusion as Intervention

Diffusion of Innovations

- **The network is not observable**
- **Only public information is observable**

Diffusion of Innovation

- An innovation is "*an idea, practice, or object that is perceived as new by an individual or other unit of adoption*"

- The theory of diffusion of innovations aims to answer **why** and **how** innovations spread.

- It also describes the **reasons** behind the diffusion process, individuals involved, as well as the rate at which ideas spread.

Innovation Characteristics

For an innovation to be adopted, the individuals adopting it (adopters) and the innovation must have certain qualities

Innovations must:

- **Be Observable**,
  - The degree to which the results of an innovation are visible to potential adopters
- **Have Relative Advantage**
  - The degree to which the innovation is perceived to be superior to current practice
- **Be Compatible**
  - The degree to which the innovation is perceived to be consistent with socio-cultural values, previous ideas, and/or perceived needs
- **Have Trialability**
  - The degree to which the innovation can be experienced on a limited basis
- **Not be Complex**
  - The degree to which an innovation is difficult to use or understand.

Diffusion of Innovations Models



- First model was introduced by Gabriel Tarde in the early 20th century

I. The Iowa Study of Hybrid Corn Seed

- Ryan and Gross studied the adoption of hybrid seed corn by farmers in Iowa
  - The hybrid corn was highly resistant to diseases and other catastrophes such as droughts

- Despite the fact that the use of new seed could lead to an increase in quality and production, the adoption by Iowa farmers was slow
  - Farmers did not adopt it due to its high price and its inability to reproduce
    - i.e., new seeds have to be purchased from the seed provider

I. The Iowa Study of Hybrid Corn Seed

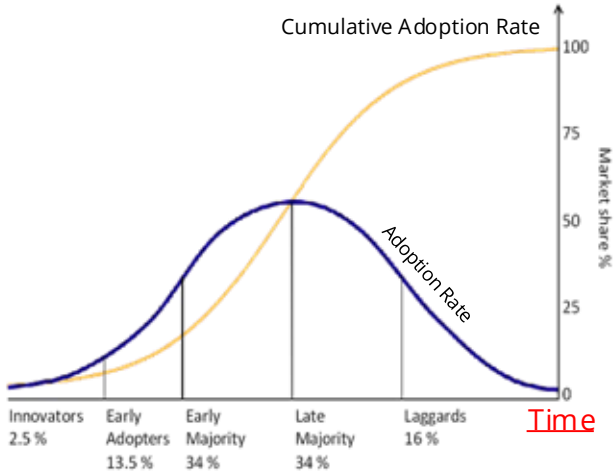
Farmers received information through two main channels

- **Mass communications** from companies selling the seeds (i.e., information)
- **Interpersonal communications** with other farmers. (i.e., influence)

- Adoption depended on a combination of information and influence.

- The study showed that the adoption rate follows an *S*-shaped curve and that there are 5 different types of adopters based on the order that they adopt the innovations, namely:

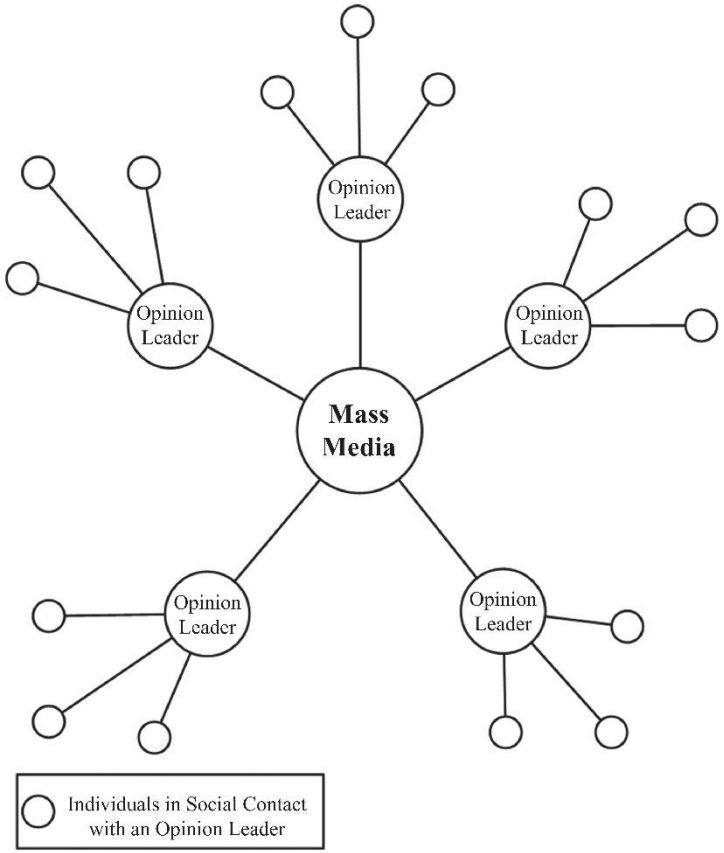
1) **Innovators** (top **2.5%**)
2) **Early Adopters** (next **13.5%**)
3) **Early Majority** (next **34%**)
4) **Late Majority** (next **34%**)
5) **Laggards** (last **16%**)



II. Katz Two-Step Flow Model

- **Two-step Flow Model.** most information comes from mass media, which is then directed toward influential figures called *opinion leaders*.

- These leaders then convey the information (or form opinions) and act as hub for other members of the society



Task 1: Two-Step Flow Model - Influence of Opinion Leaders

- **Question**: In a network of 200 individuals, 20 opinion leaders receive information from mass media. Each opinion leader has a probability of 0.7 of influencing their followers, and each leader has an average of 5 followers. Calculate the expected number of followers who receive the information.

III. Rogers: Diffusion of Innovations: The Process

Adoption process:

1. **Awareness**
   - The individual becomes aware of the innovation, but her information regarding the product is limited
2. **Interest**
   - The individual shows interest in the product and seeks more information
3. **Evaluation**
   - The individual tries the product in his mind and decides whether or not to adopt it
4. **Trial**
   - The individual performs a trial use of the product
5. **Adoption**
   - The individual decides to continue the trial and adopts the product for full use

Task 2: Diffusion of Innovations - Adoption Process

- **Question**: Suppose there is a new product launched in a market of 1000 potential customers. Each stage of the adoption process has a probability as follows: Awareness (0.8), Interest (0.6), Evaluation (0.5), Trial (0.4), and Adoption (0.3). Calculate the expected number of people who will adopt the product.

Modeling Diffusion of Innovations

The diffusion of innovations models can be concretely described as

$$\frac{dA(t)}{dt} = i(t)[P - A(t)]$$

- $A(t)$ is the total population that adopted the innovation
- $i(t)$ denotes the coefficient of diffusion corresponding to the innovativeness of the product being adopted
- P is the total number of potential adopters (till time t)

- The rate depends on how innovative the product is
- The rate affects the potential adopters that have not yet adopted the product.

Diffusion Models

Three models of diffusion:
i.e., each having different ways to compute i(t):

$$\frac{dA(t)}{dt} = i(t)[P - A(t)]$$

$$i(t) = \alpha, \quad \text{External-Influence Model}$$
$$i(t) = \beta A(t), \quad \text{Internal-Influence Model}$$
$$i(t) = \alpha + \beta A(t). \quad \text{Mixed-Influence Model}$$

- α **- Innovativeness factor of the product**
- β **- Imitation factor**

Diffusion of Innovation: Intervention

1. **Limiting the distribution of the product or the audience that can adopt the product**.

2. **Reducing interest in the product being sold.**
   - A company can inform adopters of the faulty status of the product.

3. **Reducing interactions within the population.**
   - Reduced interactions result in less imitations on product adoptions and a general decrease in the trend of adoptions.

Home Task

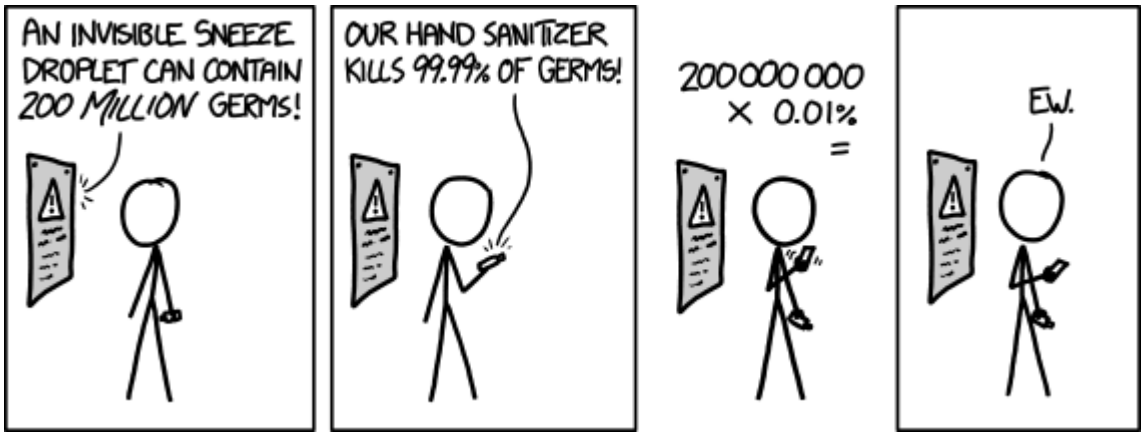
- A technology company is launching a new product in three distinct markets, each with 1000 potential adopters. They want to analyze how the product adoption will progress in each market based on different diffusion influences:

1. **Market A** follows an **external-influence model** with an innovativeness factor (α) of 0.05.
2. **Market B** follows an **internal-influence model** with an imitation factor (β) of 0.02, and starts with an initial adoption of 20 users.
3. **Market C** follows a **mixed-influence model** where both external and internal influences apply. The innovativeness factor (α) is 0.03, and the imitation factor (β) is 0.015. This market also starts with an initial adoption of 10 users.

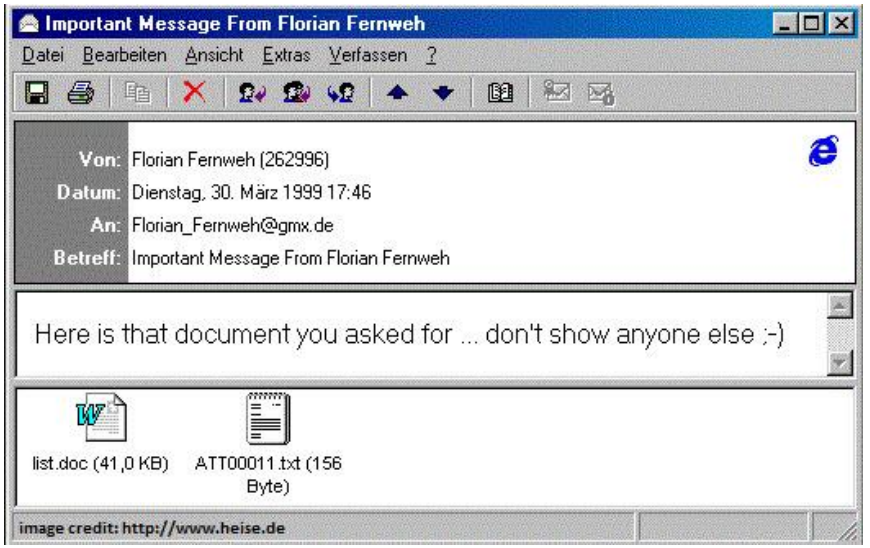
Tasks:

1. **Market A**: Calculate the expected number of adopters after 10 days.
2. **Market B**: Calculate the expected number of adopters after 5 days, assuming adoption increases each day based on the internal influence of existing adopters.
3. **Market C**: Calculate the expected number of adopters after 7 days, considering both external and internal influences.

Epidemics


AN INVISIBLE SNEEZE DROPLET CAN CONTAIN 200 MILLION GERMS!
OUR HAND SANITIZER KILLS 99.99% OF GERMS!
200000000 × 0.01% =
EW.

Epidemics: Melissa computer worm

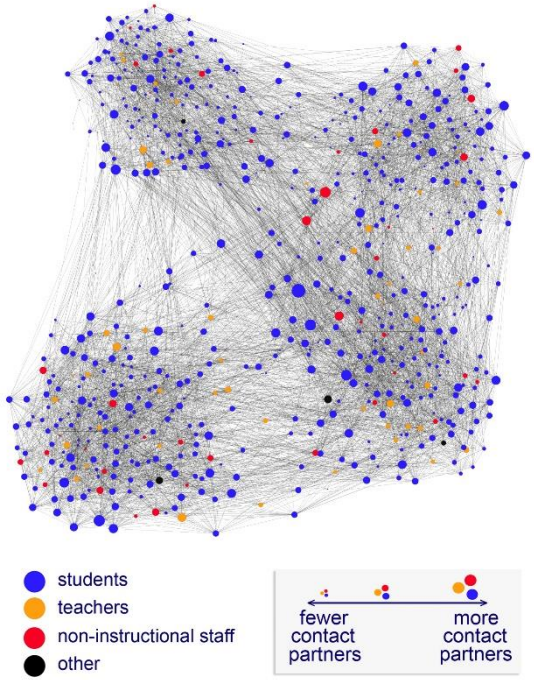


- Started on March 1999
- Infected MS Outlook users
- The user
  - Receives email with a word document that has a virus
  - Once opened, the virus sends itself to the first 50 users in the outlook address book
- First detected on Friday, March 26
  - On Monday, March 29, the virus had infected more than 100K computers

Epidemics

Epidemics describes the process by which diseases spread. This process consists of

- A pathogen
  - The disease being spread
  - Tweet being retweeted
- A population of hosts
  - Humans, animals, plants, etc.
- A spreading mechanism
  - Breathing, drinking, touching etc.



Comparing Epidemics and Cascades

- Epidemic models assume an **implicit network** and unknown connections between users.
  - Unlike information cascades and herding
  - Similar to diffusion of innovations models,

- Epidemic models are more suitable when we are interested in global patterns
  - Trends
  - Ratios of people getting infected
  - Not suitable for who infects whom

How to Analyze Epidemics?

I. Using Contact Network

- look at how hosts contact each other and devise methods that describe how epidemics happen in networks.
- **Contact network**: a graph where nodes represent the hosts and edges represent the interactions between these hosts.
  - E.g., In influenza contact network, hosts (nodes) that breathe the same air are connected

II. Fully-mixed Method

- Analyze only the rates at which hosts get infected, recover, etc. and avoid considering network information

The models discussed here will assume:

- No contact network information is available
- The process by which hosts get infected is unknown

Basic Epidemic Models

- **SI**
- **SIR**
- **SIS**
- **SIRS**

SI Model: Definition

SI model:

- The *susceptible* individuals get infected
- Once *infected,* they will never get cured

Two Types of Users:

- **Susceptible**
  - When an individual is in the susceptible state, he or she can potentially get infected by the disease.
- **Infected**
  - An infected individual has the chance of infecting susceptible parties

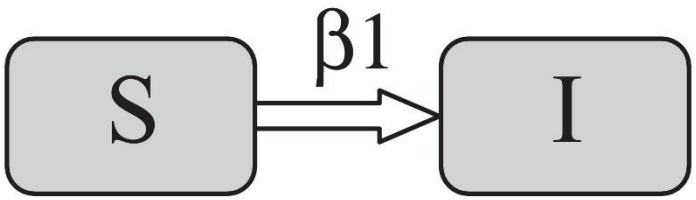
Notations

- N: size of the crowd
- $S(t)$: number of susceptible individuals at time t
  - $s(t) = S(t)/N$
- $I(t)$: number of infected individuals at time t
  - $i(t) = I(t)/N$
- β: Contact probability
  - if $\beta = 1$ everyone comes to contact with everyone else
  - if $\beta = 0$ no one meets another individual

$$N = S(t) + I(t)$$

SI Model

- At each time stamp, an infected individual will meet βN people on average and will infect βS of them
- Since I are infected, βIS will be infected in the next time step



Task 3: SI Model - Epidemic Spread

- **Question**: In a susceptible population of 500 people, 10 people are initially infected. Each infected person has a probability of 0.2 of infecting each of their 4 connections. Calculate the expected number of new infections after one round.

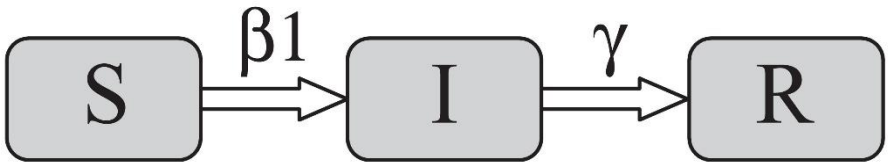
SIR Model

SIR model:

- In addition to the **I** and **S** states, a recovery state **R** is present

- Individuals get infected and some recover

- Once hosts recover (or are removed) they can no longer get infected and are not susceptible

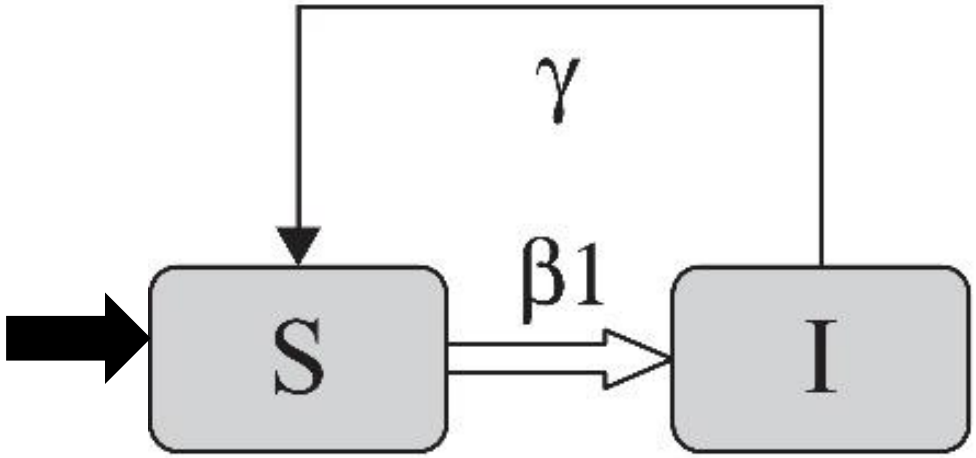


Task 4: SIR Model - Spread with Recovery

- **Question**: In a population of 600 people, 20 are infected initially, and each infected person has a probability of 0.3 of infecting each of their 3 connections. If there is a recovery probability of 0.5 per time step, calculate the expected number of people who will still be infected after one round.

SIS Model

- The **SIS** model is the same as the **SI** model with the addition of infected nodes recovering and becoming susceptible again



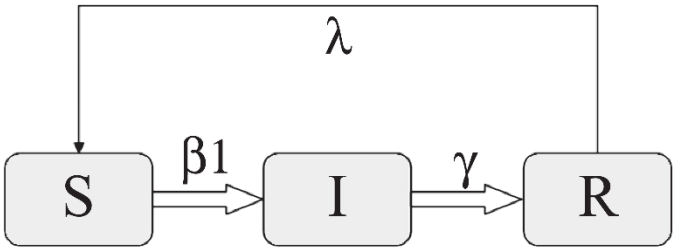
γ defines the recovering probability of an infected individual at a time stamp

Task 5: SIS Model - Recurrent Infection

- **Question**: In a network of 400 people, 30 are initially infected, and each infected individual has a probability of 0.25 of infecting each of their 4 connections. Each infected person has a probability of 0.3 of recovery, after which they become susceptible again. Calculate the expected number of infections after one round.

SIRS Model

The individuals who have recovered will lose immunity after a certain period of time and will become susceptible again



Like the SIR, model this model has no closed form solution, so numerical integration can be used

Epidemic Intervention

Epidemic Intervention

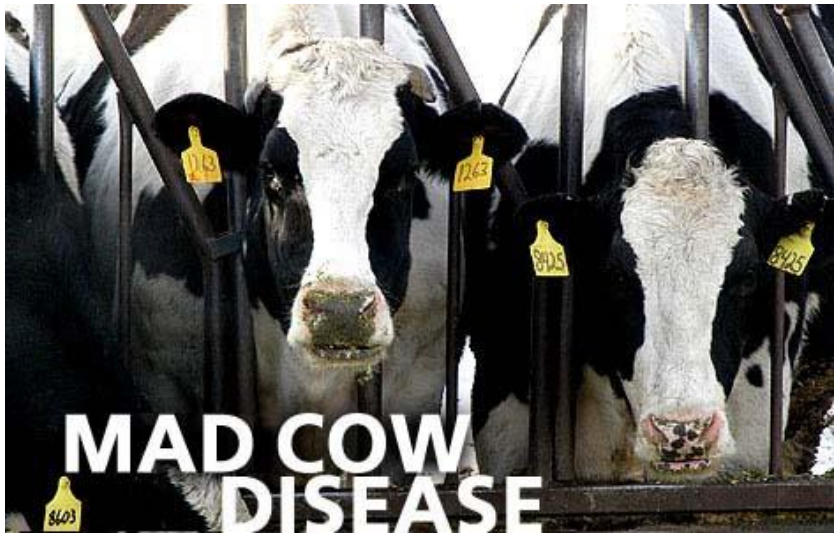
- Suppose that we have a susceptible society and want to prevent more spread by vaccinating the most vulnerable individuals

- How to find the most vulnerable individuals?

Randomly pick some nodes and ask them who is the most vulnerable from their point of view, then vaccinate those individuals!

Epidemic Intervention: *Mad Cow Disease*

- Jan. 2001
  - First case observed in UK
- Feb. 2001
  - 43 farms infected
- Sep. 2001
  - 9000 farms infected



In the mad cow disease case, we have weak ties

- Animals being bought and sold
- Soil from tourists, etc.

How to stop the disease:

- Banned movement (make contagion harder)
- Killed millions of animals (remove weak ties)